

ACRME SKU-Level Placement WHAT-IF — User Guide (Phase 2)

Workbook: `acrme_sku_placement_whatif.xlsx`

Builder: `build_acrme_sku_scoring_whatif.py`

Baseline: Azure Capacity & Quota Management Consolidated Requirements Baseline v2.4

Status: Mockup / feasibility — derived planning figures only; not connected to live Azure state. Does **not** modify the v2.4 baseline or the Phase 1 workbook.

1. What this is (and how it differs from the region-selection what-if)

This is a **new, standalone** what-if — a different kind of analysis from `acrme_region_selection_whatif_acrme_regions.xlsx`.

	Region-selection what-if	This — SKU-level placement what-if (Phase 2)
Capacity grain	ONE blended number per region	Per SKU (CAP-022/023)
Quota grain	ONE number per environment/region	Per VM/quota family (QUA-002/003/004)
Availability term (α)	blended region free-pool	per-SKU <code>GROUP AVAIL = MIN(reserved-free, family quota-available)</code>
Question answered	“Which region overall?”	“For this multi-SKU workload, which region for Prod / CVAL / DR — and is it READY for every SKU?”
Readiness	region-level	RDY-002 per candidate (all active SKU lines must clear)
SKUs per run	single	up to 8 SKU lines per run

Phase 2 is the design-doc step “availability-by-group in scoring: point α at per-SKU group-availability; emit RDY-002 readiness per candidate.” It is **read-only** — the freeze & logical-lock layer is **Phase 3** and is not in this file.

2. Workbook structure (processing order)

Order	Sheet	Role	Editable
1	ReadMe	In-workbook overview	No
2	Request	The placement request	Yellow cells
3	Policy	Score weights & thresholds	Yellow cells
4	SKU_Family_Map	SKU → quota family lookup	No
5	SKU_Catalogue	Managed SKU set (vCPU, family, flags)	No
6	Capacity_By_SKU	Reservation plane: Region × Env × SKU	No (rebuild)
7	Quota_Groups	Quota plane: pooled group per Region × Family	No (rebuild)
8	Score_Prod	SKU-grain placement score — Production	No — formulas
9	Score_CVAL	SKU-grain placement score — CVAL / Non-Prod	No — formulas
10	Score_DR	SKU-grain placement score — Disaster Recovery	No — formulas
11	Result	Recommended Prod/ CVAL/DR + ranked candidates	No — formulas

Data flow: Request + Policy → (read against) Capacity_By_SKU + Quota_Groups → Score_Prod / Score_CVAL / Score_DR → Result . Every computed cell is a live formula; edits propagate instantly.

3. How to run a scenario

Request sheet (yellow cells only)

Cell	Input	Notes
B4	Geography	Dropdown: <code>US</code> or <code>EU</code> . Filters eligible regions.
B5	Customer ID	Free text — flows to the Result sheet header.
B8-B15	SKU (per line)	Dropdown of the 8 managed SKUs. Leave blank to skip a line.
C8-C15	VM Count (per line)	Positive integer. Leave blank to skip a line.

Both **SKU** and **VM Count** must be filled for a line to be active. Leaving either blank removes that line from all scoring.

Derived (auto-filled, read-only):

Column	Derivation
D — vCPU / instance	<code>VL00KUP(SKU, SKU_Catalogue)</code>
E — Line Cores	<code>VM Count × vCPU</code>
F — Quota Family	<code>VL00KUP(SKU, SKU_Family_Map)</code>

B17 (Active SKU lines) and **B18 (Total requested cores)** are summary counters — read-only.

Policy sheet (optional tuning)

Tune weights `w_alpha` / `w_beta` / `w_epsilon` (must sum to 1.00 — verify `B7`) and thresholds `risk_threshold` (default 0.10) and `min_buffer_floor` .

Reading the results

Go to the **Result** sheet. It shows:

- **Recommended Placement** — the top-pick region per environment (Prod / CVAL / DR), its Placement Score, and its RDY-002 readiness state. Each recommendation satisfies **every active SKU line**.
- **Overall Readiness** — a summary signal based on the Prod top-pick.
- **Ranked Prod Candidates** — all 8 regions ranked by Placement Score (only eligible regions get a rank).

4. The scoring model (per candidate region, across all active SKU lines)

A region is **eligible** only when **every active SKU line** clears the availability gate. The score aggregates across lines using **bottleneck semantics** — a single thin line drags all components down.

Availability gate (Meets-All)

For each active line i the sheet computes:

```
GROUP_AVAIL_i = MIN( SUMIFS(Capacity_By_SKU.ReservedFree, Region, SKU_i),
                     SUMIFS(Quota_Groups.QuotaAvail, Region, Family_i) )

line_meets_i = IF( GROUP_AVAIL_i >= LineCore_i, 1, 0 )
```

A blank line contributes 1 (not a constraint). The region-level gate is:

```
Meets-All = MIN( line_meets_1, ..., line_meets_8 )  [= 0 if any active line fails]
```

Eligible = 1 only when InGeo = 1 **AND** Meets-All = 1.

Score components (eligible candidates only)

Each component is the **MIN** across active lines — the tightest constraint drives the score:

Symbol	Formula per active line	Aggregate
α (availability fit)	$1 - \text{LineCore}_i / \text{MIN}(\text{ReservedFree}_i, \text{QuotaAvail}_i)$ — headroom cushion left	$\text{MIN}(\alpha_i)$
β (quota headroom)	$\text{QuotaAvail}_i / \text{QuotaLimit}_i$	$\text{MIN}(\beta_i)$
ϵ (buffer safety)	$\text{ReservedFree}_i / \text{Reserved}_i$	$\text{MIN}(\epsilon_i)$

Blank lines contribute 1 to every MIN (neutral). Components are clamped to $[0, 1]$.

Placement Score: $\text{PS} = w_{\alpha} \cdot \alpha + w_{\beta} \cdot \beta + w_{\epsilon} \cdot \epsilon$ (defaults 0.50 / 0.30 / 0.20).

Rank: dense rank of PS among eligible candidates (Rank 1 = highest PS). The **top pick** per environment is INDEX/MATCH on Rank = 1; if no region is eligible it shows NONE ELIGIBLE.

Binding Constraint: for eligible rows, shows whether CAPACITY or QUOTA is the tighter limit (i.e., which of ϵ vs β is lower across lines).

5. Readiness states (RDY-002, per candidate)

The readiness hierarchy evaluates in strict priority order:

State	Condition
NO_REQUEST	No active SKU lines (B17 = 0).
RESERVATION_DEFICIT	At least one active line has <code>ReservedFree = 0</code> for this region.
QUOTA_DEFICIT	At least one active line has <code>QuotaAvail = 0</code> for this region.
CAPACITY_DEFICIT	<code>Meets-All = 0</code> — <code>GROUP AVAIL < LineCore</code> for at least one line, but no full zero-capacity line.
READY_WITH_RISK	Eligible (<code>Meets-All = 1</code>), but <code>MIN(α_i) < risk_threshold</code> (0.10 default) — thin cushion across lines.
READY	Eligible with comfortable cushion across all lines.

The **Binding Constraint** column (col O) shows `CAPACITY` or `QUOTA` for eligible rows; `N/A` for ineligible.

6. Worked example (pre-loaded sample — verified with a headless recalc, 0 formula errors)

The workbook ships with a 2-line sample request (US geography):

Line	SKU	VMs	vCPU	Line Cores	Family
1	Standard_E32ads_v5	4	32	128 cores	memory_optimized
2	Standard_D8as_v5	2	8	16 cores	general_purpose

Total requested cores: 144 (shown in B18). **Active lines: 2** (shown in B17).

A region must have `GROUP AVAIL ≥ 128` for E32ads_v5 **and** `GROUP AVAIL ≥ 16` for D8as_v5 to pass Meets-All. The tighter constraint (E32ads_v5, 128 cores) dominates α , β , and ϵ via the MIN aggregation.

Expected observations (US, Prod):

- **East US 2** — passes both lines, `READY`. High α because E32ads_v5 headroom is large relative to 128 cores.
- **Canada Central** — mock region ($0.5 \times \text{mean}$); likely `CAPACITY_DEFICIT` or `READY_WITH_RISK` on the E32ads_v5 line.

- **West US 3** — `RESERVATION_DEFICIT` by design (greenfield, ~0 observed allocation → quota limit = 0).
- **DR** — 30% bootstrap capacity (ENV-005); thin cushions push to `READY_WITH_RISK` for the large E32ads_v5 line.

To test a single-SKU scenario, clear lines 2–8 (leave B9:C15 blank) and update line 1 as needed. The formulas automatically reduce to single-line evaluation.

7. Data provenance & rulings baked in

- **Allocated** cores are derived from real usage (`SKU_USage.xlsx`). **Canada Central** is a flagged mock ($0.5 \times$ mean of the other US regions — it is absent from the source data). **DR = 30% of Prod** (ENV-005 bootstrap, not a full duplicate).
- **West US 3** shows deficits across SKUs by design: the source data has ~0 observed allocation there, so its quota limit is 0 — the correct signal for a reserved-but-unprovisioned greenfield region.
- Rulings from Phase 1: one **pooled** quota group per region+family across Prod+NonProd+DR (QUA-004); ENV-003 hard separation applies to **capacity reservations only** (quota pooled — separate planes); quota **hoarded** to one pool and distributed on demand (QUA-003).

8. Limitations / deferred

- **Read-only feasibility** — does not modify the v2.4 baseline or the Phase 1 workbook.
 - **8-line cap** — the Request sheet supports up to 8 concurrent SKU lines. Workloads exceeding 8 SKU types require a separate run or a builder extension.
 - **CRG scope is regional** (`crg-...-reg`); per-AZ CRGs (CAP-023, `crg-...-az1`) are a later increment.
 - **Freeze & logical lock** (the `LOCKED` state + `Allocation_Ledger` , logical-vs-physical CAP-002 distinction) is **Phase 3** — not in this file.
 - **SKU→family map is naming-derived**; a canonical Azure family table would replace `SKU_Family_Map` .
 - `total_customers` (γ distribution term) remains undefined in v2.4 — not used by this model, still an open baseline issue.
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Phase 2 SKU-grain placement what-if (multi-SKU, up to 8 lines) — baseline v2.4 · updated 24 Sep 2026